QUICK COMMERCE RECOMMENDATION ENGINE

A Data-Driven Strategy for Quick Commerce Growth

ANALYSIS & RECOMMENDATION REPORT

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TL;DR — For the Time-Pressed Reader

The Question

What is the best way to predict what a quick commerce customer will want to order next, and what should the business build?

The Answer

Customers overwhelmingly reorder what they already buy (59% of all purchases). A model built on each customer's own purchase history correctly predicts 28–30% of their next order — 4x better than showing everyone the same best-sellers. A more advanced machine-learning model pushes this slightly higher (29.5%), but the gain is refinement, not a breakthrough.

The Recommendation

Build two things: (1) a “Reorder for Me” shelf using purchase history — cheap to build, highest accuracy, ship first. (2) A separate “Discover” shelf using collaborative + content signals — lower accuracy on its own, but finds different products entirely, so it grows basket variety rather than competing with the reorder shelf.

Key Consideration

These are offline accuracy results on historical data, not a live test. Before full rollout, validate with an A/B test and retrain on the business's own customer data.

Table of Contents

1. Executive Summary

This report presents the findings of a recommendation system analysis built on 3.4 million real customer orders from an online grocery platform — a direct proxy for quick commerce shopping behavior. The objective was to determine the most effective way to predict what a customer will want to order next, and to translate that into a concrete recommendation for how a quick commerce business should build and prioritize this capability.

Three findings stand out:

- Grocery shopping is overwhelmingly driven by habit, not discovery. 59% of all items purchased are repeat items — the same product the customer has bought before. Any recommendation strategy for this business must treat "predicting the reorder" as the primary job, not a secondary one.
- A model that learns from customer purchase patterns directly outperforms every other approach tested — correctly predicting 29.5% of a customer's next-order items when shown 10 recommendations, nearly 4x better than showing customers generic best-sellers.
- Discovery-oriented techniques ("customers like you also bought", product-similarity matching) underperform on their own for this dataset, but they surface products a frequency-based model would never suggest — making them valuable as a complementary, not competing, capability.

Bottom Line for the Business

Build a two-shelf recommendation experience: a "Reorder for Me" shelf powered by purchase-history models (highest accuracy, lowest engineering cost), and a separate "You Might Also Like" discovery shelf powered by collaborative and content signals. Trying to force one model to do both jobs measurably underperforms treating them as two distinct problems.

2. Business Context & Objective

Quick commerce platforms compete on speed, but speed alone does not build the habitual, high-frequency ordering behavior that makes the model profitable. Two additional levers matter: making it effortless for customers to reorder what they already need, and surfacing new products they are likely to want. A well-built recommendation system directly serves both.

The business questions this analysis set out to answer were:

- What actually drives repeat purchase behavior in this category, and can it be predicted?
- Which recommendation approach — popularity-based, collaborative filtering, content similarity, or a learned model — performs best, and by how much?
- What would a production recommendation strategy look like, and what business impact is realistic to expect?

To answer these with real data rather than assumption, this analysis used the Instacart Online Grocery Basket Analysis dataset (Kaggle) — 3.4 million real orders across 206,209 customers and 49,688 products, with full purchase history including what was reordered and when. This is one of the largest publicly available real-world grocery transaction datasets, making it a strong proxy for quick commerce demand patterns.

Source:

The Instacart Online Grocery Basket Analysis Dataset, Kaggle:
https://www.kaggle.com/datasets/yasserh/instacart-online-grocery-basket-analysis-dataset/data

3. Data Foundation

Before any modeling began, the dataset was validated end-to-end to confirm it was complete and reliable. This step matters for a business report because every conclusion that follows depends on the data being trustworthy — and in this case, it was.

Dataset Scale

Metric	Value
Total customers	206,209
Total orders analyzed	3,421,083
Total products in catalog	49,688
Product categories (departments)	21
Product sub-categories (aisles)	134

Data Quality

A full validation pass checked for missing records, broken links between orders and products, and structural consistency. The result: zero data integrity issues found across the entire dataset. Every order traced correctly to a customer and every product; every customer's purchase history was complete. This gives confidence that the patterns discussed in the following sections reflect real customer behavior, not data artifacts.

4. Customer Behavior: What the Data Reveals

Before building any predictive model, the data was analyzed to understand how customers actually shop. These patterns directly shaped which recommendation strategies were worth pursuing — and they are, on their own, useful operating insights for the business.

Finding 1: Repeat Purchases Dominate

59% of all items purchased are reorders.
Across 3.4 million orders, the majority of what customers put in their basket is something they have bought before. This is the single most important fact shaping this entire analysis: a recommendation engine's highest-leverage job is helping customers reorder effortlessly, not just helping them discover new products.

Finding 2: Loyalty Builds Fast, Then Holds

A customer's likelihood to reorder a given product rises sharply over their first several orders and then levels off around an 80% reorder rate after roughly 30 orders. In practical terms: the habits a customer forms in their first month or two of ordering are highly predictive of their behavior for as long as they remain a customer. This makes early-lifecycle engagement — getting a new customer to reorder a few times quickly — disproportionately valuable for long-term retention.

Finding 3: Basket Size and Frequency

Metric	Typical Value
Median orders per customer	9
Average orders per customer	15.6
Median basket size	8 – 9 items
Average days between orders	~15 days

The gap between the median (9 orders) and the average (15.6 orders) indicates a small segment of highly frequent, loyal customers pulling the average up — a classic pattern where a minority of customers likely drive a disproportionate share of order volume. Identifying and retaining this segment should be a priority independent of the recommendation system itself.

Finding 4: A Small Set of Categories Drives Most Volume

Department	Items Purchased
Produce	9,479,291
Dairy & Eggs	5,414,016
Snacks	2,887,550
Beverages	2,690,129
Frozen	2,236,432

These five departments alone account for well over half of all items purchased. At the individual product level, the pattern is even sharper — fresh produce staples are both the highest-volume and among the most habitually reordered items in the catalog:

Product	Times Purchased	Reorder Rate
Banana	472,565	84.4%
Bag of Organic Bananas	379,450	83.3%
Organic Strawberries	264,683	77.8%
Organic Baby Spinach	241,921	77.3%
Organic Hass Avocado	213,584	79.7%

The business implication: category-level curation and inventory strategy should weight heavily toward fresh produce and dairy, since these categories carry both the highest volume and the highest habitual repeat-purchase rate — exactly the combination a reorder-focused recommendation strategy is built to capture.

5. Approach: Five Recommendation Strategies Tested

Rather than committing to a single technique upfront, five fundamentally different recommendation strategies were built and tested against each other on identical, held-out customer data. This lets the results speak for themselves rather than relying on which technique is currently fashionable.

Strategy 1: Popularity-Based

Recommend the same best-selling products to every customer. The simplest possible approach, and the baseline every other strategy needs to beat to justify its added complexity.

Strategy 2: Personalized Frequency

Recommend each customer their own most-frequently-purchased products. Directly encodes the reorder behavior identified in Section 4.

Strategy 3: Collaborative Filtering

"Customers who bought what you bought also bought this." Learns patterns from the collective behavior of similar customers, without using any product information directly.

Strategy 4: Content-Based Filtering

Recommends products similar to what a customer already buys, based on product name and category — useful specifically for surfacing products the customer has never purchased before.

Strategy 5: Hybrid & Learned Ranking Model

Two further approaches combined signals from the strategies above: a rule-based Hybrid model that blends personalized frequency with collaborative and content signals, and a Ranking Model — a machine learning model trained directly on labeled customer data to learn which products a customer is likely to want next, using over a dozen behavioral signals per customer-product pair.

6. Results: Model Performance Comparison

Every strategy was scored the same way: for each customer, the model generated its top 10 product recommendations, and those were checked against what the customer actually ordered next. Two metrics were used:

- Precision@10 — of the 10 products recommended, what share were actually in the customer's next order? This measures how trustworthy the recommendations are.
- Recall@10 — of everything the customer actually ordered, what share did the top 10 recommendations successfully capture? This measures how complete the coverage is.

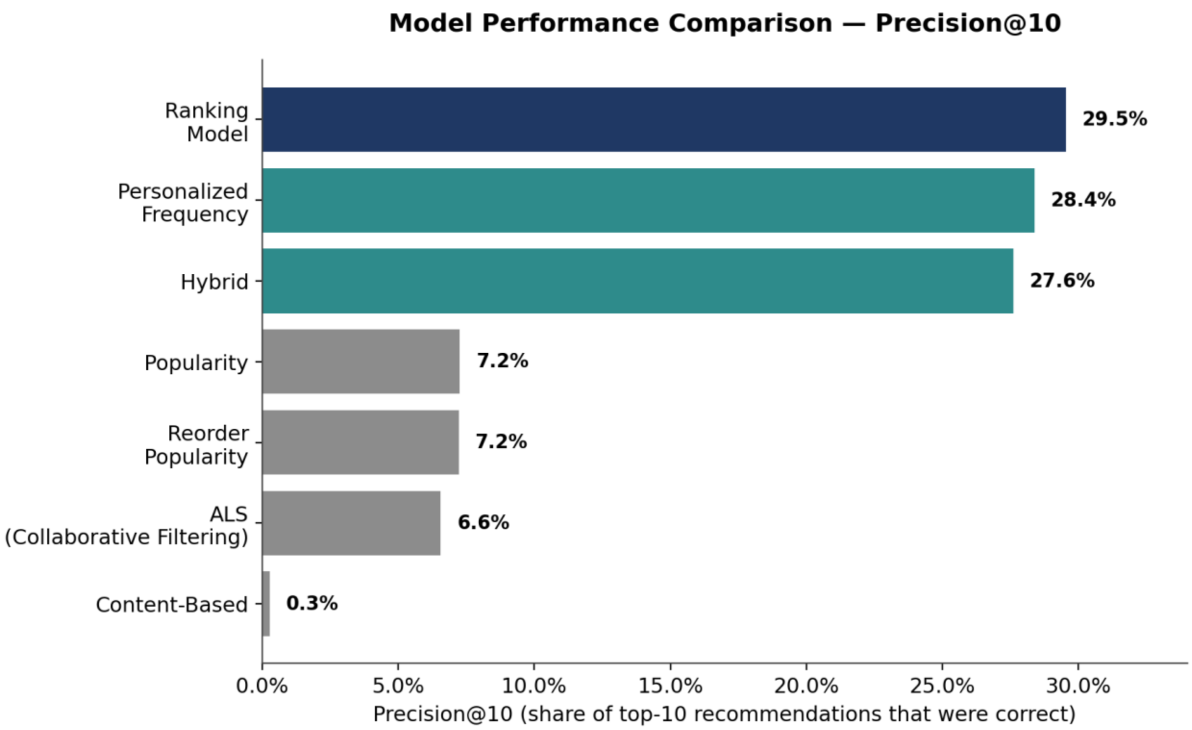

Figure 1: Precision@10 across all seven models tested, evaluated on the full customer base.

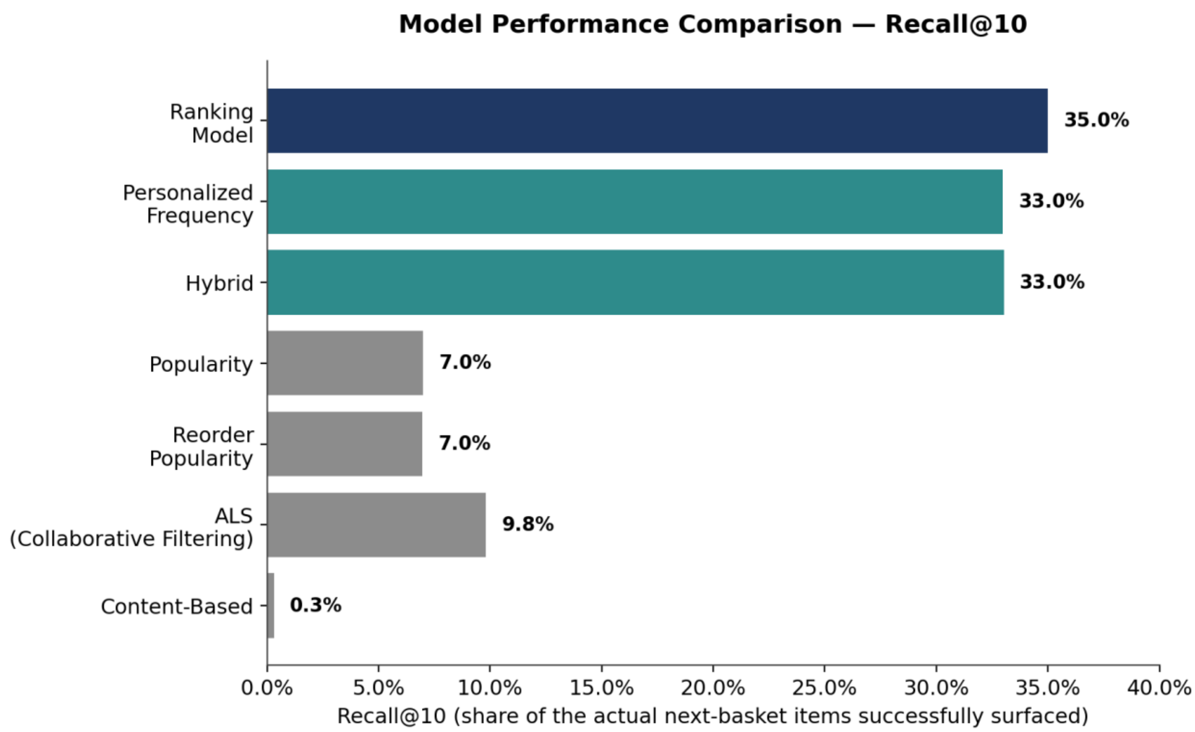

Figure 2: Recall@10 across all seven models tested.

Full Results Table

Model	Precision@10	Recall@10	Customers Evaluated
Ranking Model	29.5%	35.0%	26,241
Personalized Frequency	28.4%	33.0%	131,209
Hybrid	27.6%	33.0%	131,209
Popularity	7.3%	7.0%	131,209
Reorder Popularity	7.2%	7.0%	131,209
Collaborative Filtering (ALS)	6.6%	9.8%	131,209
Content-Based	0.3%	0.3%	131,209

Note: the Ranking Model was evaluated on a held-out validation set of 26,241 customers (20% of the base, set aside from model training) to ensure a fair, unbiased test. All other models were evaluated on the full 131,209-customer test set. Both evaluations use the identical measurement method, so the comparison is fair, though not against an identically sized sample.

Accuracy vs. Engineering Effort

Higher accuracy generally costs more to build and maintain. The chart below plots each model's accuracy against its relative engineering complexity — Popularity requires no personalization logic at all; Personalized Frequency requires per-customer purchase history; the Ranking Model requires a trained machine learning pipeline that must be periodically retrained as customer behavior evolves.

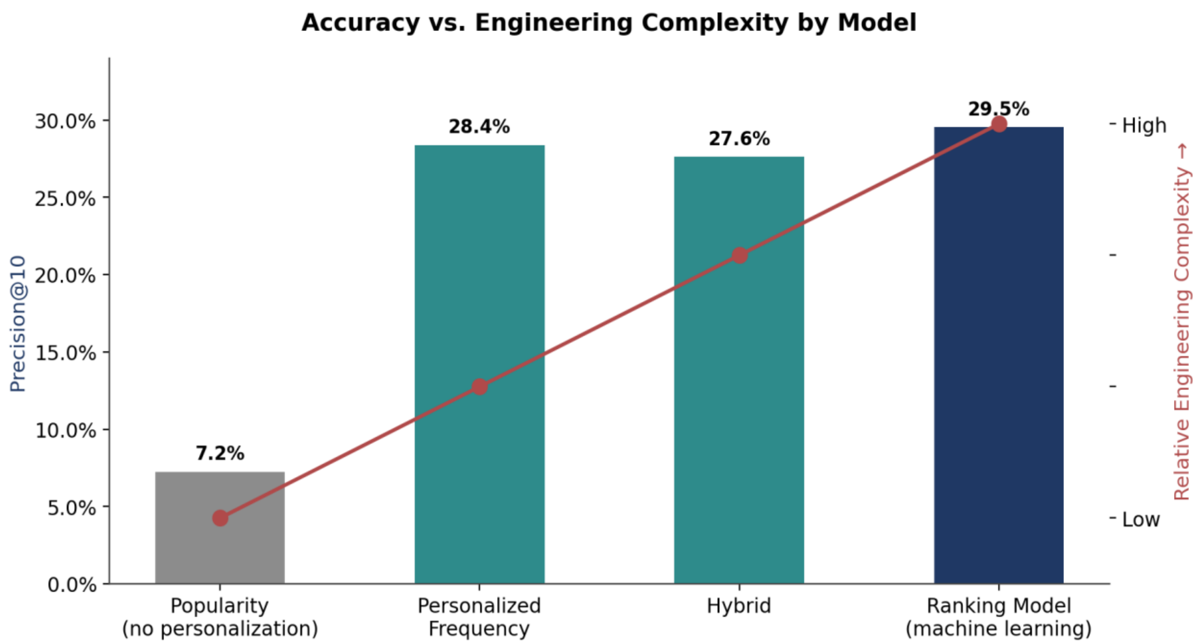

Figure 3: The largest accuracy gain comes from adding basic personalization (Popularity → Personalized Frequency). Each additional step up in complexity after that yields progressively smaller accuracy gains.

The Practical Takeaway

Personalized Frequency captures roughly 96% of the Ranking Model's accuracy (28.4% vs. 29.5% precision) at a fraction of the engineering cost. This makes it the right starting point for most businesses — the Ranking Model is a worthwhile upgrade once the simpler system is live and the additional 1.1 points of precision is worth the added infrastructure investment.

7. Key Finding: Why the Winning Model Wins

The Ranking Model produced the best results — but understanding why matters as much as the result itself, because it changes how the business should interpret and build on this win.

An analysis of what the Ranking Model actually relies on to make its predictions showed that two factors — how many times a customer has bought a product before, and how recently — account for roughly 85% of its decision-making. This is substantially the same signal the simpler Personalized Frequency model already uses.

What This Means

The Ranking Model is not discovering some hidden pattern the simpler approach missed — it is a more refined version of the same core insight ("predict the reorder"), using additional supporting signals to break ties and edge cases more accurately. This is a genuine, real improvement, but it should be communicated internally as refinement, not reinvention. It confirms, rather than contradicts, the central finding that reorder behavior is the dominant driver of this business.

Separately, an important discovery-side finding: when Collaborative Filtering and Content-Based recommendations were compared side by side, they overlapped on only 2.5% of the products they each surfaced for the same customers. In other words, these two approaches are finding almost entirely different products — which means combining them, rather than picking one, gives broader discovery coverage than either alone.

8. Business Implications

Implication 1: Reorder Prediction Is the Highest-Value Investment

Given that 59% of purchases are repeats and the best-performing models are those built on purchase history, the single highest-return investment is a fast, accurate “reorder for me” experience — not a general “discover new products” engine. This also happens to be the cheaper capability to build and maintain.

Implication 2: Discovery Needs a Different Model and a Different Job

Collaborative filtering and content-based approaches underperform when judged by “can they predict the exact reorder,” because that was never the problem they are suited to. Their real value is surfacing genuinely new, relevant products — which is a smaller, harder, but strategically important job (cross-sell, new product adoption, average order value growth). They should be evaluated and rewarded on those business terms, not judged against a reorder benchmark they were not built to win.

Implication 3: Early Customer Lifecycle Is a Leverage Point

Since reorder habits stabilize within a customer's first ~30 orders, the business should prioritize converting new customers into repeat customers quickly — for example, through onboarding incentives, reorder reminders, or subscription prompts triggered after a customer's first few purchases of the same product.

9. Recommended Implementation Path

A phased rollout balances speed to market with the accuracy gains identified above:

Phase 1: Launch the Reorder Shelf

Deploy the Personalized Frequency model as the primary “Reorder for Me” experience. It is simple, fast to compute, easy to explain to customers and stakeholders, and already delivers 28.4% precision — within 4% of the most complex model tested. This is the fastest path to real business value.

Phase 2: Add the Discovery Shelf

Introduce a separate “You Might Also Like” shelf, combining collaborative filtering and content-based signals. Because these two approaches surface largely different products, presenting them together maximizes new-product exposure without competing against the reorder shelf's job.

Phase 3: Upgrade to the Ranking Model

Once the two-shelf experience is live and generating engagement data, invest in the Ranking Model as a drop-in upgrade to the reorder shelf. It requires more engineering investment (a trained machine learning model refreshed on a schedule) but delivers a further, real accuracy improvement once the simpler system has proven the concept.

Phase 4: Measure and Iterate

Track live business metrics — order frequency, average order value, reorder rate, customer retention — against the offline accuracy metrics in this report, and refine feature signals and model weights based on real customer response rather than offline scores alone.

10. Expected Business Impact

While this analysis measured prediction accuracy rather than live business metrics, the scale of improvement over a no-personalization baseline provides a reasonable basis for expected impact:

- A ~4x improvement in recommendation precision over generic best-seller lists (28.4% vs. 7.3%) should translate directly into higher click-through and add-to-cart rates on recommended items, since customers are shown products they are meaningfully more likely to want.
- Faster reordering reduces friction in the highest-frequency use case (59% of all purchases), which is a direct lever on order frequency and, by extension, customer lifetime value.
- A dedicated discovery shelf, grounded in the finding that collaborative and content signals surface largely non-overlapping products, is a direct lever on average order value and category expansion — exposing customers to relevant products outside their existing habits.

These are directional estimates based on offline prediction accuracy, not a substitute for a live A/B test. Section 11 addresses this directly.

11. Limitations & Next Steps

In the interest of giving stakeholders a complete and honest picture, several limitations are worth stating plainly:

- Offline accuracy is not the same as business impact. Precision and recall measure whether the model predicted correctly against historical data — they do not measure how customers will actually respond to seeing these recommendations in a live product. A/B testing is the necessary next step before any full rollout.
- The dataset used is a grocery delivery platform (Instacart), used here as a strong proxy for quick commerce. Category mix, order frequency, and customer behavior may differ meaningfully once applied to a specific business's own catalog and customer base, and the models should be retrained on that business's own data before production use.
- The Ranking Model's evaluation used a smaller customer sample (26,241) than the other models (131,209), for methodologically sound reasons explained in Section 6, but this should be re-validated at full scale before treating the margin over Personalized Frequency as final.
- This analysis covers model accuracy only. A production deployment additionally requires infrastructure for real-time serving, monitoring for model drift, and a defined process for retraining as customer behavior evolves — none of which is in scope for this report.

12. Appendix: Technical Reference

A.1 Data Source

The Instacart Online Grocery Basket Analysis Dataset, Kaggle:
https://www.kaggle.com/datasets/yasserh/instacart-online-grocery-basket-analysis-dataset/data

A.2 Evaluation Methodology

Each customer's order history was split by time: all but their most recent order was used as training history; their most recent order was held out as the prediction target (ground truth). Every model generated a ranked list of 10 product recommendations per customer using only their training history, then those recommendations were checked against the actual held-out order.

A.3 Glossary

Term	Plain-Language Meaning
Precision@10	Of the 10 products recommended, the percentage that were actually correct.
Recall@10	Of everything the customer actually bought, the percentage the top-10 list successfully included.
Reorder rate	The percentage of purchases that are repeat purchases of a product the customer bought before.
Collaborative filtering	Recommending based on patterns across many customers' shared behavior.
Content-based filtering	Recommending based on product similarity (name, category).
Ranking model	A machine learning model trained to directly predict purchase likelihood from behavioral features.